

Test Design Document

****Course:**** Software Testing and Validation

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1. Equivalence Partitioning (EP) and Boundary Value Analysis (BVA)

Variable 1: Customer Age

****Rules**:** Must be ≥ 21 . If 21-24, \$20 surcharge. If ≥ 25 , standard rate.

Partition	Valid/Invalid	Boundaries to Test	Test Cases Derived
Age < 21	Invalid	20	TC_Age_01: Age 20 (Expected: Error)
21 ≤ Age < 25	Valid	21, 24	TC_Age_02: Age 21 (Expected: Surcharge applied) TC_Age_03: Age 24 (Expected: Surcharge applied)
Age ≥ 25	Valid	25, 30 (nominal)	TC_Age_04: Age 25 (Expected: Standard rate) TC_Age_05: Age 30 (Expected: Standard rate)

Variable 2: Rental Duration (Days)

****Rules**:** Must be ≥ 1 and ≤ 30 .

Partition	Valid/Invalid	Boundaries to Test	Test Cases Derived
Days ≤ 0	Invalid	0 and a negative value	TC_Dur_01: 0 days (Expected: Error); TC_Dur_05: -1 day (Expected: Error)
1 ≤ Days ≤ 30	Valid	1, 7, 8, 30	TC_Dur_02: 1 day (Expected: Success) TC_Dur_06: 7 days (Expected: no long-rental discount) TC_Dur_07: 8 days (Expected: long-rental discount) TC_Dur_03: 30 days (Expected: Success)
Days > 30	Invalid	31	TC_Dur_04: 31 days (Expected: Error)

2. Decision Table Testing

****Logic**:** Discount Calculation based on Premium Status and Duration.

* Condition 1 (C1): Is Premium Member? (T/F)

* Condition 2 (C2): Duration > 7 days? (T/F)

Conditions / Actions	Rule 1	Rule 2	Rule 3	Rule 4
C1: Premium Member?	True	True	False	False
C2: Duration > 7 Days?	True	False	True	False
Action: Discount %	20%	5%	10%	0%

****Test Cases Derived:****

* **TC_DT_01** (R1): Premium=True, Days=10. Expected: 20% discount.
* **TC_DT_02** (R2): Premium=True, Days=5. Expected: 5% discount.
* **TC_DT_03** (R3): Premium=False, Days=10. Expected: 10% discount.
* **TC_DT_04** (R4): Premium=False, Days=5. Expected: 0% discount.

3. State Transition Testing

****System State**:** Booking Lifecycle.

****Valid States**:** `REQUESTED`, `CONFIRMED`, `ACTIVE`, `RETURNED`, `CANCELLED`.

Current State	Input / Event	Next State	Expected Output	Test Case
REQUESTED	Approve Booking	CONFIRMED	State saved as CONFIRMED	TC_ST_01
REQUESTED	Cancel Booking	CANCELLED	State saved as CANCELLED	TC_ST_02

CONFIRMED	Pick up Vehicle	ACTIVE	State saved as ACTIVE	TC_ST_03
CONFIRMED	Cancel Booking	CANCELLED	State saved as CANCELLED	TC_ST_04
ACTIVE	Return Vehicle	RETURNED	State saved as RETURNED	TC_ST_05
CANCELLED	Approve Booking	N/A (Invalid)	InvalidStateTransitionError	TC_ST_06

4. Acceptance/UAT Scenarios

ID	User scenario	Expected result	Automated evidence
UAT-01	Customer books an available vehicle with age 28 for three days.	Booking is created and a confirmation message is shown.	<code>`rental/tests/test_system.py::test_successful_booking_journey`</code>
UAT-02	Customer submits age 19.	Booking is rejected and the age error is visible.	<code>`rental/tests/test_system.py::test_failed_booking_journey_underage`</code>
UAT-03	Customer submits a duration outside 1 to 30 days.	Booking is rejected with a duration error.	Unit and view tests